İlker Kayra Orman

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Profile Summary

B.Sc. graduate and M.Sc. student in Electrical Engineering, currently in the final course of completing a dual degree in Control and Automation Engineering. Expertise in reinforcement learning (RL), computational modeling, and industrial decarbonization. Experienced in Python-based predictive tools and heavy-industry energy audits at Siemens. Demonstrated leadership in international sustainability policy forums, dedicated to advancing AI-driven renewable energy transitions.

Education

Istanbul Technical University — Istanbul, Türkiye
Master of Science in Electrical Engineering — 09/2026 – Ongoing

Istanbul Technical University — Istanbul, Türkiye
Bachelor of Science in Electrical Engineering — 09/2021 – 06/2026
CGPA: 3.15 / 4.00
Senior Thesis: Design of a Permanent Magnet Linear Synchronous Motor Drive and Field-Oriented Control
Designed gate-drive circuits and current sensing hardware; implemented Field-Oriented Control and SVPWM algorithms for dynamic trajectory tracking and precise motor regulation.
Graduate-Level Coursework: Power Electronics Systems, Applied Statistics and Data Analytics in Energy, Energy Efficient Building Design
Relevant Coursework: Electric Machines, Power Electronics, Electrical Power Generation, Power Distribution, Transmission Lines, High Voltage Engineering

Istanbul Technical University — Istanbul, Türkiye
Bachelor of Science in Control and Automation Engineering (Double Major) — Expected Graduation: 01/2027
CGPA: 3.14 / 4.00
Senior Thesis: Energy Optimization in Wind Turbines using Reinforcement Learning
Developed a high-fidelity Python simulation environment; synthesized control and adaptive optimization algorithms to outperform traditional MPPT.
Relevant Coursework: Feedback Control Systems,Computer Controlled Systems, Optimal Control, State-Space Controller Design, System Modelling and Simulation

Work Experience

Siemens Digital Industries – Energy Efficiency Services — Istanbul, Türkiye
Part-Time Working Student – Energy Efficiency Engineer — 07/2024 – 07/2026

- Conducted on-site industrial audits in Iron & Steel, Chemical, and MDF plants; performed sensor instrumentation, electrical power quality data logging, and closed-loop process efficiency evaluations.
- Collaborated with Siemens EMEA/China teams; designed, wired, and commissioned PLC/sensor hardware test rigs to upskill engineering teams on standardized industrial measurement protocols.
- Developed 3 Python/Excel tools for dynamic load forecasting and energy optimization; integrated field measurement datasets with PowerBI dashboards for technical root-cause analyses.

ITU Entrepreneurship and Innovation Center (GINOVA) — Istanbul, Türkiye
Part-Time Student Assistant – Project Support — 09/2023 – 02/2025

- Coordinated technical documentation, milestones, and cross-functional communication among 7 international partners for the EU-funded "Technological Well-being" project.
- Managed logistics and data tracking for 20+ deep-tech and sustainability events (2,000+ participants); facilitated entrepreneurship training programs for engineering teams.

Pubinno Inc. — Istanbul, Türkiye
Embedded Software Intern — 01/2023 – 03/2023

- Designed and programmed embedded electronic test fixtures and validation units for two commercial hardware platforms.
- Implemented C-based real-time algorithms for sensor data acquisition and anomaly detection, improving subsystem reliability in IoT-enabled fluid systems.
- Enhanced industrial communication robustness (UART/SPI/Modbus protocols) and packet integrity across distributed device architectures.

Research

ITU Advanced Vehicle Technologies Autonomous and Power Systems Research Lab. Istanbul, Türkiye
Undergraduate Research Assistant 09/2024 – Present

- Overhauled and restored an inactive linear PMSM testbench by troubleshooting mechanical wear, re-aligning components, and renewing outdated electronic hardware.
- Modernized motor control software and refurbished driver units to bring the experimental drive system back to operational capability.
- Designed, fabricated, and validated a high-power PMSM motor driver unit, implementing field-oriented control (FOC) algorithms.
- Coordinated an international Research-Based Learning project funded by the EELISA alliance, hosting and mentoring 10 international students from 3 partner universities in lab workflows.

Book Chapter Contributor: Net Zero — Decarbonising the Global Economies 2024
Chapters: Global Energy Supply, Transportation — Publisher: Springer

- Contributed research on decarbonization of the global energy supply and transportation sectors; supervisor: Dr. Sinan Küfeoğlu.

International and Additional Training

ISU Energy Summer School
International Summer University on Renewable Energy Participant with Mobility Grant (4 ECTS) 09/2026

Tsinghua University Department of Electrical Engineering
Future Power and Energy Systems Summer School Participant with Mobility Grant 08/2026

Vocational Qualification Authority (Türkiye)
Industrial Electric Maintenance Technician — Level 4
Certificate Code: YB0356/22UY0496-4/00/209 10/2024

Project Management Institute
Certified Associate in PMI-ITU CEC 10/2023

TU Delft, DelftX
EnergyX: Designing a Sustainable Energy and Renewable Future (Certificate Link) 06/2024

Sant'Anna School of Advanced Studies
Seasonal School: Economics of Innovation and Technological Change 10/2023

Madrid Polytechnic University & TotalEnergies
Climate Change and Energy Transition 06/2023

Czech Technical University
Lean Six Sigma Management — *Verification Code: TUK5-DRRX-2UPV-78GG* (Verification Link) 09/2023

University Politehnica of Bucharest
Introduction to Smart Grids — *Verification Code: GKAA-FATW-NWU3-G4E2* (Verification Link) 09/2022

Social and Policy Engagement

European Parliament – European Student Assembly (ESA) Strasbourg, France
Policy Contributor 2025

- Co-authored two adopted policy recommendations on institutional decarbonization frameworks.

European Engineering Learning Innovation Alliance (EELISA) 2024 – 2026

- Co-chaired the Student Council and coordinated multi-partner technical initiatives, securing €40,000+ in EU funding supported by Siemens Germany and municipal partners.

Volunteering

Türkiye Technology Team Foundation (T3) 09/2024 – 05/2025
Volunteer STEM Mentor — Mentored student teams on robotics and engineering projects for national competitions.
Bilim Virüsü & Enerjisa 01/2023 – 05/2024
Volunteer Instructor — Delivered interactive workshops on energy efficiency and sustainability to 100+ students.

Skills

Modelling & Tools: MATLAB/Simulink, TIA Portal, Python, KiCad, PowerBI, Excel, Closed-Loop Regulation, PID Tuning, FOC, Instrumentation, Automation, Microcontrollers, Algorithm Design, System Modelling
Languages: Turkish (Native), English (C1 — IELTS 7.5), Chinese (HSK1/HSK2 — Beginner).
Interests: Camping, Skiing, Cooking